

Collective epithelial migration mediated by the unbinding of hexatic defects

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eLife Assessment

This **important** theoretical study shows that active hexatic topological defects in epithelia enable collective cell flows. Within the general limitations of coarse-grained hydrodynamic models in fully capturing cell-scale behavior, the study provides **compelling** evidence supporting its conclusions. These findings will be of interest to both biophysicists studying collective cell behaviors and biologists investigating epithelial flows during development.

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Abstract

Collective cell migration in epithelia relies on *cell intercalation*: a local remodelling of the cellular network that allows neighbouring cells to swap their positions. Unlike foams and passive cellular fluid, in epithelial intercalation these rearrangements crucially depend on activity. During these processes, the local geometry of the network and the contractile forces generated therein conspire to produce a burst of remodelling events, which collectively give rise to a vortical flow at the mesoscopic length scale. In this article we formulate a continuum theory of the mechanism driving this process, built upon recent advances towards understanding the hexatic (i.e. 6-fold ordered) structure of epithelial layers. Using a combination of active hydrodynamics and cell-resolved numerical simulations, we demonstrate that cell intercalation takes place via the unbinding of topological defects, naturally initiated by fluctuations and whose late-times dynamics is governed by the interplay between passive attractive forces and active self-propulsion. Our approach sheds light on the structure of the cellular forces driving collective migration in epithelia and provides an explanation of the observed extensile activity of *in vitro* epithelial layers.

Introduction

From humble soap froths (Weaire and Hutzler, 1999 [↗](#); Graner et al., 2008 [↗](#); Marmottant et al., 2008 [↗](#)) down to epithelial layers, confluent cellular fluids use intercalation to flow, even in the absence of gaps and interstitial structures. At the heart of the process is a mechanism known as *topological rearrangement process of the first kind*, or T1 for brevity, through which the vertices of a honeycomb network merge and then split, thereby leading to a remodelling of the network's topology. In epithelia, this strategy allows cells to migrate, while preserving the structural integrity of the tissue, hence its major biological functionalities. These include the barrier function, which allows epithelial layers to maintain homeostasis, ensure nutrient transport and filter out harmful pathogens, as well as extrusion and replacement of apoptotic cell (Irvine and Wieschaus, 1994 [↗](#); Keller et al., 2000 [↗](#); Walck-Shannon and Hardin, 2014 [↗](#); Tetley et al., 2016 [↗](#); Tetley and Mao, 2018 [↗](#); Paré and Zallen, 2020 [↗](#); Rauzi, 2020 [↗](#)).

While relying exclusively of local rearrangements of the cellular network, intercalation gives rise to surprisingly organized patterns, where cells are able to migrate collectively over distances orders of magnitude larger than the average cell size. In the morphogenesis of *Drosophila*, for instance, cell intercalation drives a major cellular rearrangement known as germ-band extension, in which a layer of cells initially localized in the ventral region of the developing embryo, folds over the dorsal region upon extending by approximately two and half time its initial length. In this process, individual cells persistently move across the anterior-posterior axis more than ten times their body length. In *Drosophila*, intercalation is also the main process driving the salivary gland tube formation, in which cells radially converge towards a central pit and eventually escape from the tangent plane of the embryo, thereby giving rise to the tubular structure (Sanchez-Corrales et al., 2018 [↗](#)).

In the context of cancer progression, the role of cell intercalation has been recently debated in relation with various stages of the so-called *metastatic cascade*: i.e. biomechanical pathway leading to the formation of a secondary tumor (Cheung and Ewald, 2016 [↗](#)). The latter is schematically divided into three main phases: 1) detachment of cell clusters from a primary tumor and invasion of the extracellular matrix (ECM); 2) penetration (i.e. *intravasation*), circulation and expulsion (i.e. *extravasation*) of the clusters in and from the blood stream; 3) colonization of a healthy tissue and the proliferation of a secondary tumor. Along the cascade, metastatic cells undergo multiple phenotypic switches, aimed at maximizing their chances of success and survival within the surrounding microenvironment. These, in turn, determine the cells' motility mode, which can vary from individual (e.g. amoeboid or mesenchymal) to collective (e.g. intercalation-based or flocking guided by a small number leader cells localized at the front of the cluster) (Friedl and Gilmour, 2009 [↗](#); Haeger et al., 2019 [↗](#); Serra-Picamal et al., 2012 [↗](#); Murugan et al., 2024 [↗](#)).

Yet, while being highly regulated at the biochemical level (Cavey et al., 2008 [↗](#); Yamada et al., 2005 [↗](#); Yonemura et al., 2010 [↗](#); Buckley et al., 2014 [↗](#); Engl et al., 2014 [↗](#); Zallen and Wieschaus, 2004 [↗](#); Bertet et al., 2004 [↗](#)), cell intercalation cannot be separated from the mechanical forces originating it and whose nature, spatial organization and dynamics are still largely unknown. In this article, we provide a topological insight into the mechanics of cell intercalation, by leveraging on recent advances toward deciphering orientational order in epithelial layers (Li and Ciamarra, 2018 [↗](#); Pasupalak et al., 2020 [↗](#); Durand and Heu, 2019 [↗](#); Armengol-Collado et al., 2023a [↗](#),b [↗](#); Eckert et al., 2023 [↗](#); Cislo et al., 2023 [↗](#)). The latter originates from the cells' anisotropic shape and results in the emergence of liquid crystal phases collectively known as *p*-atics, with *p* an integer reflecting the symmetry of the system under rotation by $2\pi/p$. The honeycomb structure of epithelial layers, in particular, has been shown to give rise to *hexatic* order (i.e. $p = 6$) at length scales ranging from one to dozens of cells, depending on the cells' density and molecular repertoire, as well as the mechanical properties of the substrate (Armengol-Collado et al.,

2023a [b](#); Eckert et al., 2023 [b](#)). In the language of liquid crystals, the cell-wide morphological transformations underlying T1 processes can be described in terms of topological defects known as *disclinations*: i.e. point-like singularities in the otherwise regular configuration of a continuous p -atic order parameter - i.e. $\Psi_p = \langle e^{ip\vartheta} \rangle$, with ϑ the orientation of the individual building blocks and (...) the ensemble average (Giomi et al., 2022b [a](#)) - around which the average cellular orientation rotates by $2\pi s$, with $s = \pm 1/p, \pm 2/p \dots$ the winding number or “strength” of the defect. In passive two-dimensional matter, disclinations mediate the transition from solid to liquid via a process known as Kosterlitz-Thouless-Halperin-Nelson-Young (KTHNY) melting scenario (Kosterlitz and Thouless, 1972 [b](#), 1973 [b](#); Nelson and Halperin, 1979 [b](#); Young, 1979 [b](#); Kosterlitz, 2016 [b](#)). According to this, the hierarchical unbinding of neutral defect complexes - i.e. for which $\sum_i s_i = 0$ - renders the system progressively more disordered. Our working hypothesis is that the competition between active and passive forces drives a similar unbinding mechanism in epithelial layers. In the following, we clarify the various steps and possible outcomes of this process and test this hypothesis against both hydrodynamic and cell-resolved numerical simulations.

Results

Cell intercalation and T1 cycle

A typical cell intercalation is illustrated in **Fig. 1a** [b](#) for a cluster of four cells, hereafter referred to as *primary* cell cluster. A T1 occurs when the junction that connects the internal 3-fold coordinated vertices shrinks until they merge into a 4-fold vertex, and then split once more along the orthogonal direction. Such an *internal* T1, however, leaves the number and, importantly, the positions of *external* vertices of each cell unchanged. Thus, despite this concept having received little attention in the literature (see, e.g., Refs. (Staple et al., 2010 [b](#); Fletcher et al., 2014 [b](#); Rauzi, 2020 [b](#); Duclut et al., 2022 [b](#); Sknepnek et al., 2023 [b](#); Jain et al., 2023 [b](#))), it is impossible to achieve collective migration by means of *isolated* T1 processes.

A full cell intercalation then consists of a burst of internal and a peripheral T1 processes, as is schematically summarized in **Fig. 1a** [b](#). These steps do not necessarily occur in a sequential order, but are most often simultaneous. Furthermore, since collective migration is a phenomenon that occurs roughly homogeneously across the entire cell layer, there is no single initial T1, but a uniform distribution of seeds. Crucially, internal T1 processes do *not* always trigger a full intercalation. After the first T1, where the internal 3-fold vertices shrink to a 4-fold vertex, the cluster may reverse its dynamics and return to its original configuration (see **Fig. 1b** [b](#)). In the following, we refer to this scenario as *T1 cycle*: i.e. a direct T1 followed by an inverse T1, which does not permanently alter the configuration of the honeycomb network. Intuitively, and as we show next, whether the initial T1 triggers a full intercalation, hence collective migration, or a cycle, is determined by the configuration of contractile stresses exerted by the cells, which, in turn, are modulated by the local geometry of the primary cluster.

As a starting point, we focus on the intermediate configuration of primary cell cluster comprising two orthogonal pairs of pentagonal and heptagonal cells, as shown in the central column of **Fig. 1** [b](#). This configuration, which corresponds to the most elementary short-ranged excitation of a honeycomb network, can be described in the language of topological defects as a quadrupole of $\pm 1/6$ disclinations. As in two-dimensional melting, such a defect-structure can arise spontaneously, as consequence of spatiotemporal fluctuations of physical or biological nature. The two complementary remodelling events following the intermediate configuration, i.e. cell intercalation (**Fig. 1a** [b](#)) and the T1 cycle (**Fig. 1b** [b](#)), correspond instead to the two possible “fates” of this initial excitation. Upon unbinding [Figs. **1a**-(i) and **1b**-(i) [b](#)], the four defects comprising the quadrupole can further unbind into two $\pm 1/6$ disclination dipoles [**Fig. 1a**-(ii) [b](#)], or annihilate [**Fig. 1b**-(ii) [b](#)], thereby restoring the initial defect-free configuration. As we demonstrate next, the former scenario corresponds to cell intercalation and the latter to the T1 cycle. To this end, we

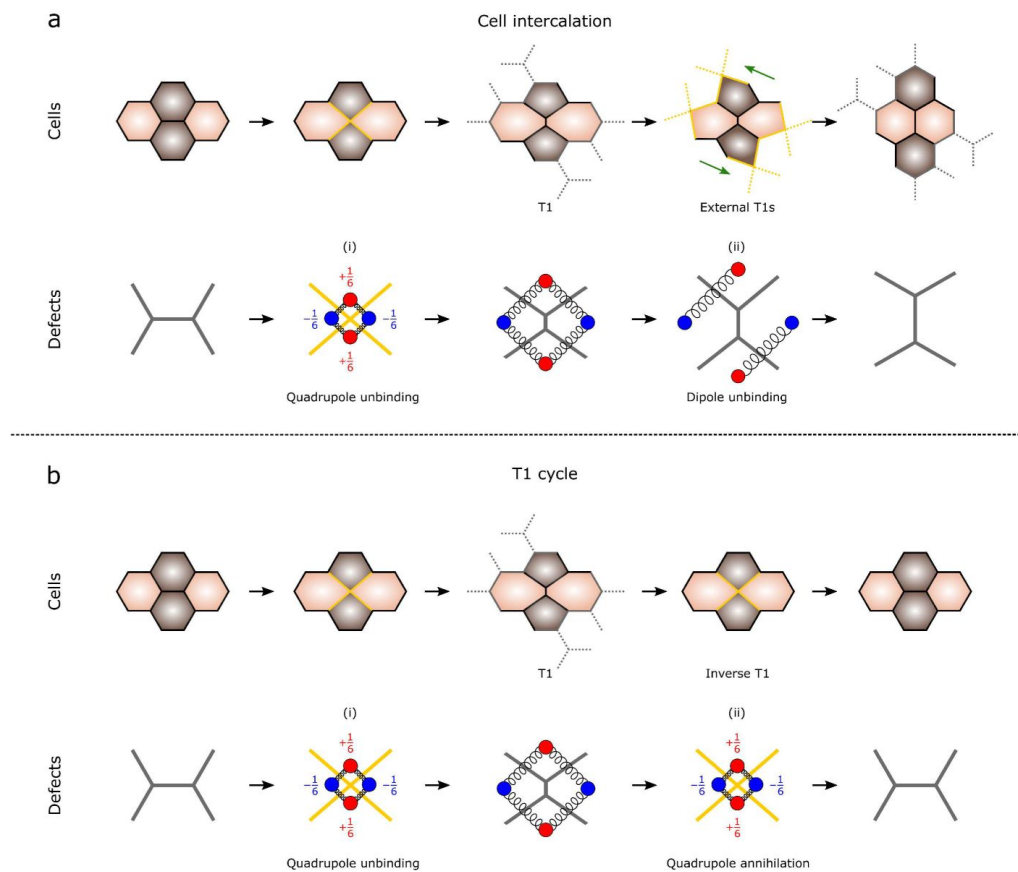


Figure 1.

Cell intercalation and T1 cycle.

(a) A full cell intercalation, consists of an *internal* and four *external* T1 processes. The latter reconfigure the peripheral vertices of the primary cluster, thereby triggering new T1 processes across the neighboring cells. In the language of topological defects, the T1 translates to the (i) unbinding of a $\pm 1/6$ defect quadrupole and (ii) a further unbinding of the quadrupole into a pair of dipoles. These two processes are schematically presented in a specific temporal order, but, in practice, they occur simultaneously or nearly so. **(b)** In a T1 cycle, the primary cell cluster undergoes a T1, followed by an inverse T1, which restores its initial configuration. The process corresponds to (i) the unbinding of a defect quadrupole and (ii) its annihilation.

identify three *geometric* requirements that a model of cell intercalation must fulfill, regardless of the desired level of biophysical accuracy. 1) The average orientation of the cells must rotate by $\pi/6$ with respect to initial configuration. 2) Both the primary cluster and its surrounding cells, must perform a local *convergent extension*: i.e. move inward in one direction, and outward along the orthogonal one (Keller et al., 2008 [DOI](#); Blanchard, 2017 [DOI](#); Wang et al., 2020 [DOI](#); Iorati-Uba et al., 2023 [DOI](#)). We stress that the adjective *local*, is used here to distinguish this process from convergent extension as intended in developmental biology, where the same rearrangement occurs at the scale of the entire organism. 3) In order to remodel the external vertices and initiate cell intercalation, the primary cluster must undergo a spontaneous shear deformation.

In the following, we show that our construction not only fulfills these requirements, but, harnessing the predictive power of active hydrodynamics, provides readily testable experimental predictions. To this end, we numerically integrate the hydrodynamic equations of active hexatic liquid crystals, introduced by Armengol-Collado *et al.* in Ref. (Armengol-Collado et al., 2023b [DOI](#)) (see Methods). To follow the fate of the primary cell cluster after an internal T1, we assume the cells to be initially horizontally oriented, so that the phase θ of the hexatic order parameter, $\Psi_6 = |\Psi_6| e^{6i\theta}$, is $\theta = 0$, and construct a configuration featuring a quadrupole of $\pm 1/6$ disclinations [see **Figs. 2a-(i) and 2b-(i)** [DOI](#)]. Along one *full* loop encircling each of these defects, θ changes by $\pm\pi/3$, with the sign reflecting that of the defect's winding number. Since the external boundary of the primary cluster consists of four *half* loops, this implies that θ varies in the range $-\pi/6 \leq \theta \leq \pi/6$ around the quadruple, as indicated by the alternating blue and red tones in **Fig. 2a-(ii) and 2b-(ii)** [DOI](#). Once the defect quadrupole breaks into two dipoles, this new orientation propagates from the boundary of the primary cluster into the space between the dipoles [see **Fig. 2a-(iii)** [DOI](#)], while leaving the orientation of the cells in the exterior essentially undistorted. Thus, the unbinding of a $\pm 1/6$ defect quadrupole from a defect-free configuration and its break up into two dipoles drives a $\pi/6$ rotation of the cells between the dipoles [see **Fig. 2a-(iv)** [DOI](#)], consistently with our first requirement. Conversely, if defects annihilate [see **Fig. 2b-(iii)** [DOI](#)], the cells' initial orientation is restored after a transient orientational fluctuation [see **Fig. 2b-(iv)** [DOI](#)].

In order to address the second and third requirements, we look at the configuration of the velocity field $\mathbf{v}$, corresponding to the average velocity of the cells in the surroundings of the primary cluster. As well documented in the theoretical (Giomi et al., 2014 [DOI](#); Giomi, 2015 [DOI](#); Hoffmann et al., 2020 [DOI](#)) and experimental (Saw et al., 2017 [DOI](#); Kawaguchi et al., 2017 [DOI](#); Blanch-Mercader et al., 2018 [DOI](#); Balasubramaniam et al., 2021 [DOI](#); Yashunsky et al., 2022 [DOI](#)) literature of active nematic liquid crystals, the distortion induced by topological defects drives a flow, whose structure and direction is determined by the defect's strength and the magnitude of the active stresses collectively exerted by the cells. Immediately after unbinding, an approximated expression for the velocity of the flow caused by the defect quadrupole can be analytically calculated (see Methods). Calling $\mathbf{r} = |\mathbf{r}|(\cos \phi \mathbf{e}_x + \sin \phi \mathbf{e}_y)$ the distance from the center of the primary cluster and ℓ the cluster's size, i.e. the distance between the center of the central junction and the center of any cell in the cluster, this velocity is given by

$$\mathbf{v} \approx \frac{120\alpha_6\ell^4}{\eta} \left[\left(4 - 6 \log \frac{|\mathbf{r}|}{\ell} - 3 \frac{|\mathbf{r}|^2}{\ell^2} \right) \cos 6\phi \right] \frac{\mathbf{r}}{|\mathbf{r}|^8}. \quad (1)$$

The constant α_6 has dimensions of force over volume, embodying the active stresses exerted by the cells and modulated by the local hexatic order, and η the shear viscosity (see Methods for details). Thus, in close proximity the primary cluster, where $|\mathbf{r}|/\ell \gtrsim 1$, Eq. (1) [DOI](#) gives $v_x(x, 0) \sim \alpha_6/(\eta\ell^4) x$ and $v_y(y, 0) \sim \alpha_6/(\eta\ell^4) y$. In agreement with our second requirement, this is typical structure of a *stagnation flow*, whose realization at the cellular scale is a local convergent extension. We stress that, consistently with the cooperative and mesoscale nature of cell intercalation, Eq. (1) [DOI](#) cannot be obtained from the mere superposition of the flows individually sourced by the defects, as a consequence of the non-linear dependence of the order parameter on

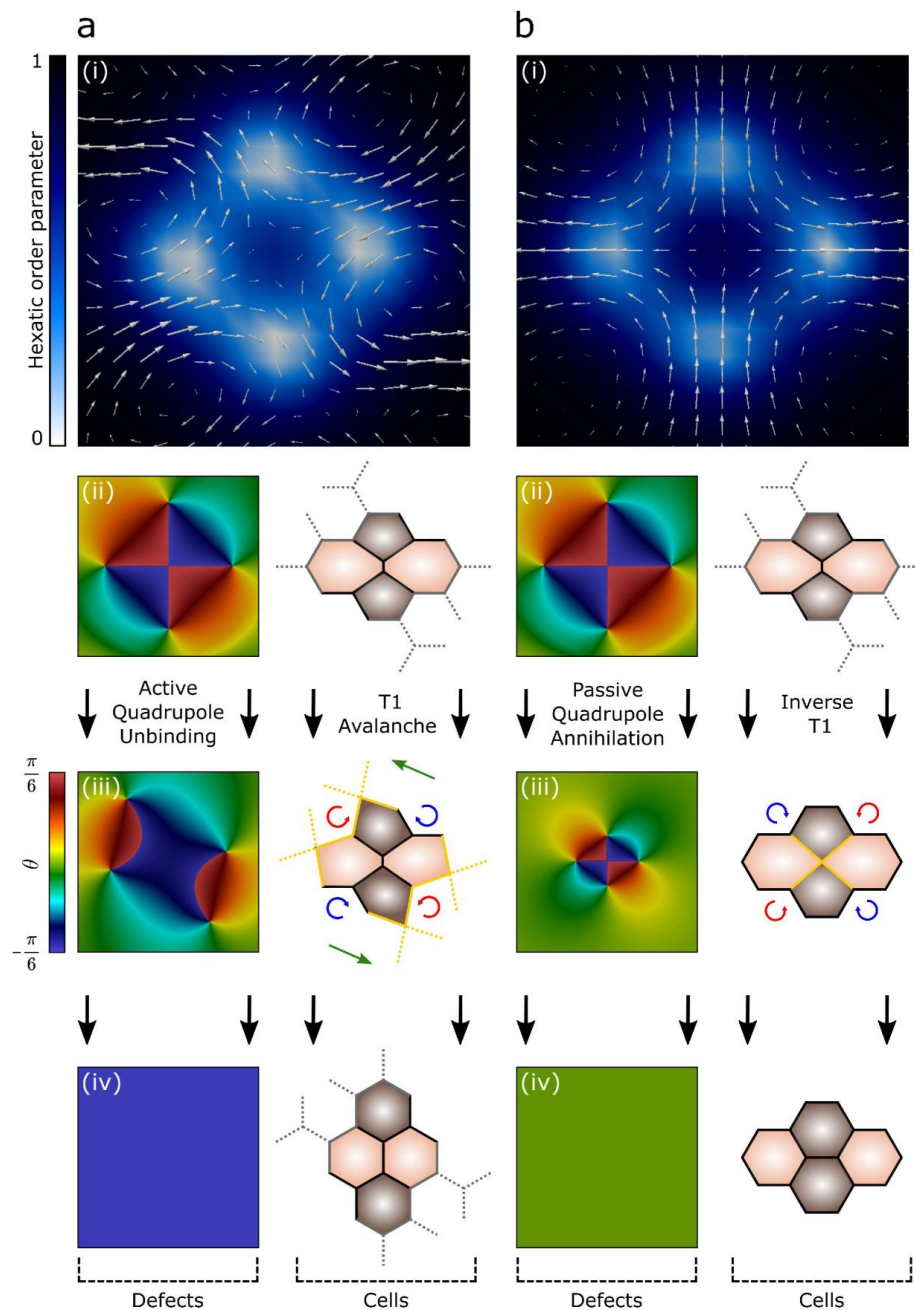


Figure 2.

Cell intercalation and T1 cycle as defect unbinding and annihilation.

(a) Cell intercalation. (i) Backflow velocity field generated during the unbinding of an *active*, hexatic defect quadrupole. The three panels below show the orientation field associated with (ii) the quadrupole in the initial configuration, (iii) as it unbinds in a pair of $\pm 1/6$ dipoles and (iv) after the dipoles have moved outside of the region of interest, together with the corresponding configuration of the primary cluster. As the dipoles move away from each other, the cells surrounding the primary cluster rotate clockwise (blue) and counterclockwise (red). **(b)** T1 cycle. (i)-(iv) Analogous sequence as in panel (a), but associated with the annihilation of the defect quadrupole. Notice that, in panel (iii), the direction of the flow is reversed. The details of the finite difference simulations can be found in Methods.

the average cellular orientation. Finally, we notice that the specific direction of motion - i.e. the sign of v - depends solely on α_6 , which, in turn, is either positive or negative depending on whether the active stresses exerted within the cell layer are respectively *contractile* or *extensile*.

As the cellular layer starts remodelling and the defects comprising the quadrupole migrate away from their original position, a solution of the hydrodynamic equations becomes analytically inaccessible, but can be obtained from a numerical integration of the hydrodynamic equations and is displayed in **Figs. 2a and 2b**, for two different α_6 values. When α_6 is large and *negative*, the quadrupole splits into two $\pm 1/6$ dipoles moving away from each other at an angle of approximately $5\pi/6$ (see Methods for details). Such a shear deformation is further enhanced by the coupling between hexatic order and flow, which, in a way not dissimilar to flow alignment effects in nematic liquid crystals, drives a rotation of the local orientation (de Gennes and Prost, 1993). This biases the unbinding dynamics of the defect dipoles, thereby setting, in concert with the passive Coulomblike forces at play, the direction along which the dipoles move away from each other (Krommydas et al., 2023). Finally, switching off active stresses - i.e. $\alpha_6 = 0$, shown in **Fig. 2b** - suppresses both defect unbinding and shear. Consistently, inverting the direction of active forces from extensile to contractile - i.e. $\alpha_6 > 0$, see Methods - results in a speed up of defects annihilation, hence of the T1 cycle, but never lead to the unbinding of $\pm 1/6$ pairs, thus to the onset of cell intercalation.

We conclude this section by stressing that the scenario emerging from our hydrodynamic analysis, of which **Eq. (1)** represents a central outcome, is further corroborated by recent numerical work by Erdemci-Tandogan *et al.* (Erdemci-Tandogan and Manning, 2021) and Das *et al.* (Das et al., 2021), who, using a different cell-resolved model of epithelia - i.e. the Vertex model (Honda and Eguchi, 1980) - showed that the rate of T1 processes enhances the fluidity of tissues. Consistently, the speed of the stagnation flow sourced by cell intercalation increases like η^{-1} , indicating that, the faster cells intercalate, the smaller η , the more fluid is the epithelial layer.

Collective Cell Migration

Having demonstrated the viability of our hydrodynamic approach, we next outline a number of general predictions, the most striking of which is that collective epithelial migration, as it results from cell intercalation, is a process of activity-guided defect unbinding. To this end, we perform numerical simulations of the multiphase field model (MPF) (Loewe et al., 2020), which have been proved to capture various aspects of epithelial organization, including the recently found hexanematic multiscale order (Armengol-Collado et al., 2023a,b; Eckert et al., 2023). In this approach, cells are modeled as droplets of immiscible fluid phases, undergoing a persistent random walk. Each cell is characterized by a nominal propulsion speed v_0 , equal for all cells, and a fluctuating direction of motion, with rotational diffusion coefficient D_r . A detailed description of the MPF model and numerical details is provided in Methods. In order to test the correlation between topological defects, T1 processes and collective migration, we fix the speed v_0 and the total number of cells and vary the persistence length of the cells trajectories by tuning the rotational diffusion coefficient D_r . **Fig. 3a,b** show a typical configuration of MPF simulations, colored according to the normalized hexatic and nematic longitudinal stress, respectively, experienced by cells in the tissue. Note that the negative value of the longitudinal stress corresponds to extensile stresses (see the Methods section for details).

To track topological defects, we first polygonize the cells by detecting their contour and mark the vertices, as shown in **Fig. 3c**. For each cell, we then compute the *shape function*

$$\gamma_6 = \frac{\sum_{v=1}^V |r_v|^6 e^{6i\phi_v}}{\sum_{v=1}^V |r_v|^6}, \quad (2)$$

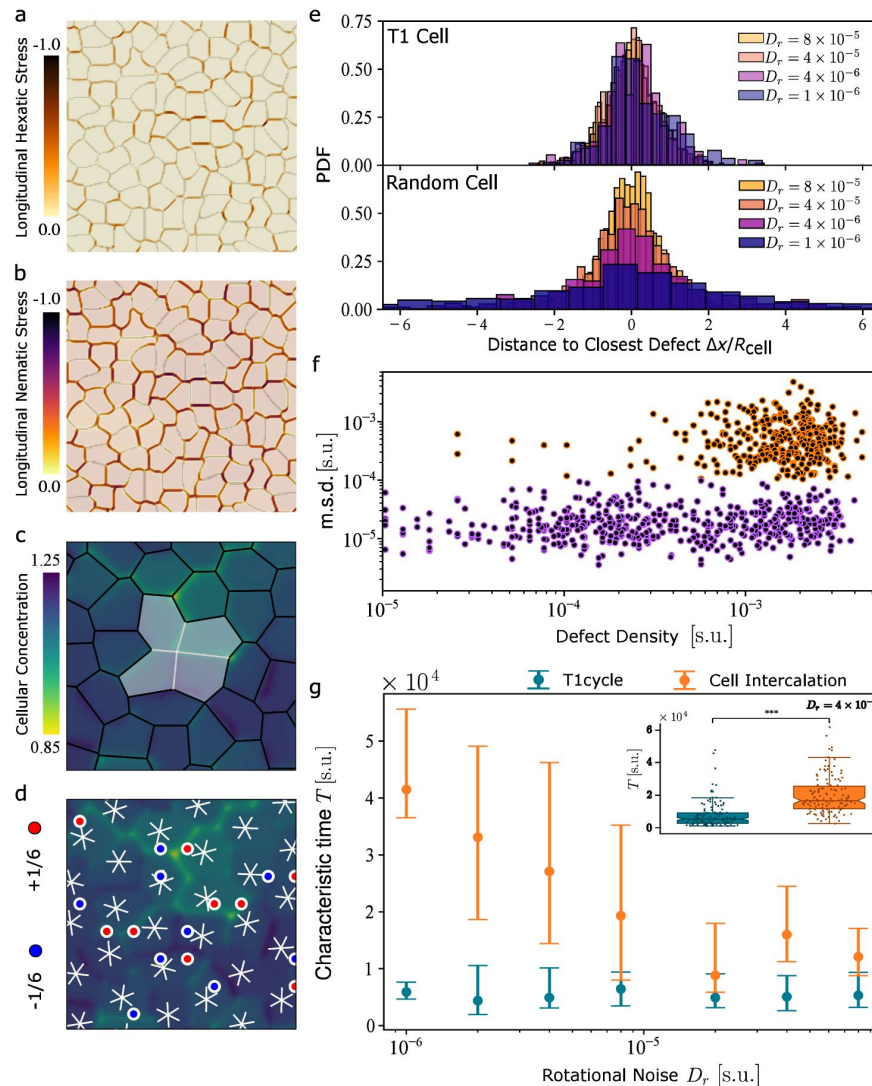


Figure 3.

Collective cell migration as defect unbinding in the multiphase field model.

(a-b) Color plots illustrating the longitudinal hexatic (a) and nematic (b) stresses in MPF simulations (refer to Methods). The color bar is normalized to the largest stress magnitude observed in the configuration. Notably, the stress is uniformly negative, reflecting the extensile characteristics of both hexatic and nematic stresses. (c) Example of a four-cell cluster as it undergoes a T1 process, together with (d) the reconstructed 6-fold orientation field. The 6-legged stars mark the local 6-fold orientation of the cells (see Methods), while the red and blue dots denotes the $+1/6$ and $-1/6$ defects. For such four-cell cluster in real epithelial cell monolayer, please see Ref. (Armengol-Collado et al., 2023a). (e) Probability distribution of finding a T1 (red tones) and a random cell (yellow tones) at a given distance from a defect, for four different values of the rotational noise D_r . The data indicate a prominent correlation between T1 process and topological defects. (f) The mean square displacement (m.s.d) of cells versus defect density computed over a time window of $\Delta t = 25 \times 10^3$ iterations, chosen to match the typical duration of T1 events and defect lifetimes. We identify two distinct sub-populations of cells: “slow” (blue tones), with no correlation to the local density, and “fast” (yellow tones), located where the local defect density is higher. The former correspond to cells undergoing a T1 cycle and the latter participating to cell intercalation, hence to collective cell migration. (g) Temporal statistics of tissue remodelling events in multiphase field simulations. Average time between two intercalation events (orange) and average period of a T1 cycle (green) versus the rotational diffusion coefficient D_r . The box plot in the inset shows the statistics of events analyzed for the case at $D_r = 4 \times 10^{-5}$. (Pairwise comparisons was performed with the two-sided t-test: $***p < 10^{-3}$). In the main graph error bars are reported as the first (bottom bar) and third (upper bar) quartile of the dataset.

where $\mathbf{r}_v$, with $v = 1, 2 \dots V$ is the positions of the v -th vertex with respect to the centroid of the polygon and $\phi_v = \arctan(y_v/x_v)$ its orientation with respect to the x -axis. As demonstrated in Ref. (Armengol-Collado et al., 2023b), the phase $\text{Arg}(\gamma_6)/6$ of the shape function identifies the 6-fold orientation of a cell and is represented as white six-legged stars in Fig. 3d. The shape function is then coarse-grained over the length scale $R = 1.5R_{\text{cell}}$ to reconstruct the *shape parameter* $\Gamma_6 = \langle \gamma_6 \rangle_R$. Finally, topological defects are identified as singularities in the otherwise smoothly varying field $\Gamma_6 = \Gamma_6(\mathbf{r})$. T1 processes, conversely, are readily detected by tracking those events leading to a recombination of the change of neighbors of a given cell (see Figs. 3c).

For each D_r value, we reconstruct the probability of finding a T1 at a given distance from a defect (see Figs. 3c and 3d) and compare it with that of an arbitrary cell. Both distributions are approximately Gaussian and have vanishing mean if expressed in terms of the signed distance Δx (see Fig. 3e). Furthermore, for larger D_r values, where the short correlation time renders the system more disordered and dense of defects, the two probability distributions are hardly distinguishable, with the probability of finding a defect in proximity of a T1 only slightly larger than that associated with an arbitrary cell. As D_r is decreased also the density of defects decreases and the latter probability distribution becomes flatter and flatter, while the former remains unchanged, thereby confirming that T1 processes are *de facto* a realization of hexatic defect unbinding. To correlate structure and dynamics we show in Fig. 3f the mean squared displacement of T1 cells as a function of local defect density. These measurements were taken over a time interval $\Delta t = 2.5 \times 10^4$, which aligns with the characteristic duration of T1 events in our simulations (see Methods section *Multiphase field model of epithelial tissues* for further details). Our data confirm the existence of two different classes: i.e. “slow” and “fast” cells, respectively denoted by purple and orange tones. While for both classes of cells the mean squared displacement is roughly uniform for all values of the local defect density, fast cells only appear where the density is higher, thus providing an alternative signature of cell intercalation and T1 cycles. Cells undergoing a T1 cycle oscillate about their initial position, but do not participate to collective migration and, therefore, exhibit small mean square displacement. By contrast intercalating cells drive collective migration and, based on the correspondence identified here, can only be found in regions of high defect densities. Note that the difference in the mean square displacement of cells undergoing intercalation is at least one order of magnitude larger than that of cells affected by an isolated T1 cycle (Fig. 3f). Hence, although isolated internal T1’s (and by extension T1 cycles) can have small long-ranged effects, those effects are negligible in comparison the collective long-ranged motion induced by a full cell intercalation.

Finally, following Refs. (Das et al., 2021; Erdemci-Tandogan and Manning, 2021) we study the temporal correlations of T1 cycles and cell intercalation. Fig. 3g shows a plot of the average time between two intercalation events and the average period of T1 cycles, for varying rotational diffusion coefficients D_r . The former displays a decreasing trend with D_r , indicating that rotational noise improves the performance of cell intercalation thus favoring a faster collective migration. Conversely, the dynamics of T1 cycles is essentially unaffected by rotational diffusion. Both behaviors can be readily rationalized on the basis of our hydrodynamic description. As previously observed in relation to Fig. 3e, rotational noise favors the proliferation of hexatic defects, hence the intercalation rate. Conversely, being defect annihilation driven by the passive attractive forces resulting from the entropic elasticity of the hexatic phase, once a T1 cycle is initiated, its duration is essentially independent on the noise strength.

Before concluding, we discuss three especially striking aspects of collective cell migration highlighted by our approach and amenable to experimental scrutiny. First, from the active flow given by Eq. (1) and neglecting irrelevant inertial effects, one can estimate the magnitude of the active forces at play: i.e. $F_{\text{active}} \sim \alpha_6/\ell^3$, where ℓ denotes the range of the distortion caused by the unbinding dislocations. Such a scaling form results primarily from the 6-fold structure of cellular forces under the assumption - still to be verified in hexatic epithelial layers, but consistent with previous observations on nematic cell cultures (Duclos et al., 2016; Saw et al., 2017);

Kawaguchi et al., 2017 [↗](#); Blanch-Mercader et al., 2018 [↗](#); Yashunsky et al., 2022 [↗](#)) - that α_6 is, at least approximately, spatially uniform. Second, in order for these forces to prompt defect unbinding, thereby triggering cell intercalation, they must overcome the passive Coulomb-like forces driving their annihilation. At the length scale of the primary cluster, the magnitude of the latter is roughly given by $F_{\text{passive}} \sim L_6/\ell$, with L_6 the orientational stiffness of the hexatic phase (see Methods). Equating F_{active} and F_{passive} provides an estimate of the typical size of the primary cluster at which the cellular layer becomes unstable to collective migration: i.e. $\ell_6 = \sqrt{|\alpha_6|/L_6}$. This active hexatic length scale was identified in Ref. (Armengol-Collado et al., 2023b [↗](#)) and is believed to play a role analog to that of the active nematic length scale $\ell_2 = \sqrt{L_2/|\alpha_2|}$ (Giomi, 2015 [↗](#)). The latter is the fundamental parameter controlling the collective behavior of active nematic liquid crystals and, depending on how it compares with other extrinsic and intrinsic length scales, determines the hydrodynamic stability (Duclos et al., 2018 [↗](#)) of active nematics, the distribution of the vortex area in chaotic cytoskeletal flows (Guillamat et al., 2017 [↗](#); Lemma et al., 2019 [↗](#)), the size of nematic domains in bacterial colonies (You et al., 2018 [↗](#)) and *in vitro* cultures of spindle-like cells (Blanch-Mercader et al., 2018 [↗](#)) etc. Similarly, we expect that confining epithelia at length scales smaller than ℓ_6 has the effect of suppressing cell intercalation and possibly render necessary different locomotion strategies and possibly favoring a switch to mesenchymal phenotypes. Finally, although actomyosin networks render individual cells contractile (Schwarz and Safran, 2002 [↗](#)), at a collective level, epithelial (Saw et al., 2017 [↗](#); Blanch-Mercader et al., 2018 [↗](#); Balasubramaniam et al., 2021 [↗](#)) and neural progenitor monolayers behave as an extensile system (Kawaguchi et al., 2017 [↗](#)). In agreement with this recent experimental evidence (Saw et al., 2017 [↗](#); Blanch-Mercader et al., 2018 [↗](#); Balasubramaniam et al., 2021 [↗](#)), our analysis shows that only an *extensile* activity can fuel collective migration in confluent epithelia and, leveraging on the familiar language of topological defects, it further provides a simple key to rationalizing the mechanical advantage of such a biological strategy: i.e. only extensile forces can provide the type of repulsive interactions that are necessary for defects to unbind, thus to trigger cell intercalation.

Discussion

In conclusion, we have investigated the physical mechanisms behind cell intercalation in confluent epithelial layers using analytics, and a combination of continuum and discrete modelling. After having established that cell intercalation, hence collective migration, requires a burst of correlated T1 processes, we demonstrated how the latter originates from the unbinding of neutral quadrupoles of hexatic disclinations. As in the KTHNY melting scenario (Nelson and Halperin, 1979 [↗](#)), the latter are spontaneously generated by fluctuations and can either annihilate, thereby restoring the original configuration (T1 cycles), or further unbind in two pairs of $\pm 1/6$ disclinations, decreasing the translational and orientation order and increasing its fluidity. As these pairs move away from each other by activity, they stir and shear the cellular network, thereby initiating further T1 processes, which, cooperatively, give rise to cell migration. To assess the significance of our predictions, we have tested them against numerical simulations of the Multiphase Field Model (MPF) (Loewe et al., 2020 [↗](#); Monfared et al., 2023 [↗](#); Camley et al., 2014 [↗](#); Palmieri et al., 2015 [↗](#); Peyret et al., 2019 [↗](#); Alert and Treppe, 2020 [↗](#)), finding good quantitative agreement. We stress, however, that the theoretical framework employed here is not specifically tailored to the MPF or other discrete models of epithelial layers - a choice that would entail the risk of constructing a “model of a model”, rather than a model of the physical system itself. Conversely, by taking a more generic top-down approach, our theory sheds light on the structure of the cellular forces driving collective migration in epithelia, suggests the possibility of a confinement-induced switch to mesenchymal phenotypes and provides an explanation of the observed extensile activity of *in vitro* epithelial layers (Saw et al., 2017 [↗](#); Blanch-Mercader et al., 2018 [↗](#); Balasubramaniam et al., 2021 [↗](#)).

Methods

Hydrodynamic equations of epithelial layers

Our hydrodynamic equations of epithelial layers have been given in Ref. (Armengol-Collado et al., 2023b) and account for both hexatic and nematic order, with the former being dominant at short and the latter at long length scales. Since cell intercalation occurs at small length scales, here we can ignore nematic order and focus solely on the hydrodynamics of the hexatic phase. In addition to the standard density ρ and velocity $\mathbf{v}$, this can be described in terms of the 6-fold order parameter tensor $\mathbf{Q}_6 = 4 \langle \mathbf{v}^{\otimes 6} \rangle = 4 |\Psi_6| \langle \mathbf{n}^{\otimes 6} \rangle$. In this expression $\otimes^6$ indicates the 6-fold tensorial product, $\mathbf{v} = \cos \theta \mathbf{e}_x + \sin \theta \mathbf{e}_y$, and $\mathbf{n} = \cos \theta \mathbf{e}_x + \sin \theta \mathbf{e}_y$, are respectively the fluctuating and average orientation of the hexatic building blocks and the operator $\langle \dots \rangle$ renders its argument traceless and symmetric (see Refs. (Giomi et al., 2022b, a) for a general introduction to p -atic hydrodynamics). The short scale hydrodynamic equations of the epithelial layer are then given, in the most generic form, by

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = (k_d - k_a) \rho, \quad (3a)$$

$$\rho \frac{D\mathbf{v}}{Dt} = \nabla \cdot \boldsymbol{\sigma} + \mathbf{f}, \quad (3b)$$

$$\frac{D\mathbf{Q}_6}{Dt} = \Gamma_6 \mathbf{H}_6 + 6 \langle \mathbf{Q}_6 \cdot \boldsymbol{\omega} \rangle + \lambda_6 \langle \nabla^{\otimes 4} \mathbf{u} \rangle + \bar{\lambda}_6 \text{tr}(\mathbf{u}) \mathbf{Q}_6 + \nu_6 \langle \mathbf{u}^{\otimes 3} \rangle. \quad (3c)$$

with $D/Dt = \partial_t + \mathbf{v} \cdot \nabla$ is the material derivative. In Eq. (3a), k_d and k_a the cell division and apoptosis rates, here assumed equal. For simplicity, we also assume uniform density throughout the system so that Eq. (3a) reduces to the standard incompressibility condition $\nabla \cdot \mathbf{v} = 0$. In Eq. (3b), $\boldsymbol{\sigma}$ is the total stress tensor and $\mathbf{f}$ an external body force. In Eq. (3c), $\mathbf{u} = [\nabla \mathbf{v} + (\nabla \mathbf{v})^T]/2$ and $\boldsymbol{\omega} = [\nabla \mathbf{v} - (\nabla \mathbf{v})^T]/2$, with T indicating transposition, are respectively the strain rate and vorticity tensors and entail the coupling between hexatic order and flow, with λ_6 and ν_6 material constants and $(\nabla^{\otimes n})_{i_1 i_2 \dots i_n} = \partial_{i_1} \partial_{i_2} \dots \partial_{i_n}$. Because of incompressibility, the term $\bar{\lambda}_6 \text{tr}(\mathbf{u}) \mathbf{Q}_6$ in Eq. (3c) vanishes. The tensor $\mathbf{H}_6 = -\delta F / \delta \mathbf{Q}_6$ is the hexatic analog of the molecular tensor, dictating the relaxation dynamics of the order parameter tensor toward the minimum of the orientational free energy

$$F = \int dA \left(\frac{L_6}{2} |\nabla \mathbf{Q}_6|^2 + \frac{A_6}{2} |\mathbf{Q}_6|^2 + \frac{B_6}{4} |\mathbf{Q}_6|^4 \right), \quad (4)$$

where $|\dots|^2$ is the Euclidean norm and is such that $|\mathbf{Q}_6|^2 = |\Psi_6|^2/2$. The constant L_6 is the order parameter stiffness, while A_6 and B_6 are phenomenological constants setting the magnitude of the coarse-grained complex order parameter at equilibrium: $|\Psi_6| = |\Psi_6^{(0)}| = \sqrt{-2A_6/B_6}$, when $H_{i_1 i_2 \dots i_6} = 0$.

The stress tensor figuring in Eq. (3b) can be customarily decomposed in a passive and an active contributions: i.e. $\boldsymbol{\sigma} = \boldsymbol{\sigma}^{(p)} + \boldsymbol{\sigma}^{(a)}$. The passive stress, in turn, can be expressed as $\boldsymbol{\sigma}^{(p)} = -P \mathbb{I} + \boldsymbol{\sigma}^{(v)} + \boldsymbol{\sigma}^{(e)} + \boldsymbol{\sigma}^{(r)}$, where P is the pressure and $\boldsymbol{\sigma}^{(v)} = 2\eta \langle \mathbf{u} \rangle$ the viscous stress, with η the shear viscosity. The tensor $\sigma_{ij}^{(e)} = -L_6 \partial_i \mathbf{Q}_6 \odot \partial_j \mathbf{Q}_6$ is the elastic stress, arising in response of a static deformation of a fluid patch and the symbol $\odot$ indicates a contraction of all matching indices of the two operands yielding a tensor whose rank equates the number of unmatched indices (two in this case). Finally $\boldsymbol{\sigma}^{(r)} = -\lambda_6 \nabla^{\otimes 4} \odot \mathbf{H}_6 + 3 (\mathbf{Q}_6 \cdot \mathbf{H}_6 - \mathbf{H}_6 \cdot \mathbf{Q}_6)$ is the reactive stress tensor, which embodies the conservative forces arising in response to flow-induced distortions of the hexatic orientation. The active stress tensor $\boldsymbol{\sigma}^{(a)}$ was introduced in Ref. (Armengol-Collado et al., 2023b) on the basis of phenomenological and microscopic arguments and is given by $\boldsymbol{\sigma}^{(a)} = a_6 \nabla^{\otimes 4} \odot \mathbf{Q}_6$, with a_6 a constant.

Multiphase field Model for Epithelial Tissues

The model

The multiphase field model is a cell-resolved model where each cell is described in a two-dimensional space by a concentration field $\phi_c = \phi_c(\mathbf{r})$, with $c = 1, 2, \dots, N_{\text{cell}}$, and N_{cell} the total number of cells in the systems (Loewe et al., 2020; Monfared et al., 2023). The equilibrium state is defined by the free energy $F = \int dA f$, where the free energy density f is

$$f = \frac{\alpha}{4} \sum_c \phi_c^2 (\phi_c - \phi_0)^2 + \frac{k_\phi}{2} \sum_c |\nabla \phi_c|^2 + \epsilon \sum_{c < c'} \phi_c^2 \phi_{c'}^2 + k_{\text{ad}} \sum_{c < c'} \nabla \phi_c \cdot \nabla \phi_{c'} + \sum_c \lambda \left(1 - \frac{1}{\pi \phi_0^2 R_{\text{cell}}^2} \int dA \phi_c^2 \right)^2. \quad (5)$$

The first two terms in the free energy represent a ϕ^4 theory for phase separation. The parameters $\alpha > 0$ and $k_\phi > 0$ control the nominal surface tension $\sigma = \sqrt{8k_\phi \alpha}$ and interfacial thickness $\xi = \sqrt{2k_\phi / \alpha}$, stabilizing spherical shapes in an isolated environment. This ensures the concentration field ϕ_c remains close to ϕ_0 inside the cell and vanishes outside. The term proportional to $\epsilon > 0$ enforces volume exclusion, preventing cell overlap, while the $k_{\text{ad}} > 0$ term models cell-cell adhesion, promoting tissue confluency in dense environments (Monfared et al., 2023). Additionally, the $\lambda > 0$ term constrains cell area near its nominal value πR_{cell}^2 , where R_{cell} is the preferred cell radius.

The phase field ϕ_c evolves via the Allen-Cahn equation:

$$\partial_t \phi_c + \mathbf{v}_c \cdot \nabla \phi_c = -M \frac{\delta F}{\delta \phi_c}, \quad (6)$$

where non-equilibrium effects arise from the non-equilibrium advection term involving $\mathbf{v}_c = v_0 (\cos \theta_c \mathbf{e}_x + \sin \theta_c \mathbf{e}_y)$. Here, v_0 is the constant self-propulsion speed of the c -th cell, and θ_c defines its migration direction. The angle θ_c , in turn, follows a stochastic dynamics:

$$\frac{d\theta_c}{dt} = \eta_c, \quad (7)$$

with noise η_c satisfying $\langle \eta_c(t) \eta_{c'}(t') \rangle = 2D_r \delta_{cc'} \delta(t - t')$, where D_r sets the noise diffusivity. The mobility M in Eq. (8) governs the relative strength of thermodynamic relaxation versus non-equilibrium migration.

We stress that, while multiphase field models can incorporate non-equilibrium effects in various ways, here the sole source of non-equilibrium is the self-propulsion velocity $\mathbf{v}_c$.

Here $\alpha > 0$ and $k_\phi > 0$ are material parameters which can be used to tune the surface tension $\sigma = \sqrt{8k_\phi \alpha}$ and the interfacial thickness $\xi = \sqrt{2k_\phi / \alpha}$ of isolated cells and thermodynamically favor spherical cell shapes, so that the concentration field is large (i.e. $\phi_c \approx \phi_0$) inside the cells and zero outside. The repulsive bulk term proportional to $\epsilon > 0$ captures the fact that cells cannot overlap, while the term proportional to $k_{\text{ad}} > 0$ models the interfacial adhesion between cells, ultimately favoring tissue confluency in a crowded environment (Monfared et al., 2023). The term proportional to $\lambda > 0$ forces the cells' area around its nominal value πR_{cell}^2 , with R_{cell} the preferential cell radius. The phase field ϕ_c evolves according to the Allen-Cahn equation

$$\partial_t \phi_c + \mathbf{v}_c \cdot \nabla \phi_c = -M \frac{\delta F}{\delta \phi_c}, \quad (8)$$

where $\mathbf{v}_c = v_0(\cos \theta_c \mathbf{e}_x + \sin \theta_c \mathbf{e}_y)$ is the velocity at which the c -th cell self-propels, with v_0 a constant speed and θ_c the angle defining the nominal direction of cell migration. The latter evolves according to the stochastic equation

$$\frac{d\theta_c}{dt} = \eta_c, \quad (9)$$

where η_c is a noise term with correlation function $\langle \eta_c(t) \eta_{c'}(t') \rangle = 2D_r \delta_{cc'} \delta(t-t')$ and D_r a constant controlling noise diffusivity. The constant M in Eq. (8) is the mobility measuring the relevance of thermodynamic relaxation with respect to non-equilibrium cell migration. Eq. (8) is solved with a finite-difference approach through a predictor-corrector finite difference Euler scheme implementing second order stencil for space derivatives (Carenza et al., 2019).

We have integrated the dynamical equations with a number of cells $N_{\text{cells}} = 440$ in a system of size 384×403 with periodic boundary conditions. The model parameters values used in the simulations are as follows: $\alpha = 0.2$, $k_\phi = 0.2$, $\epsilon = 0.1$, $k_{\text{ad}} = 0.005$, $\lambda = 600$, $R_{\text{cell}} = 11.5$, $v_0 = 0.006$. The noise variance D_r was varied in the range $10^{-6} \leq D_r \leq 8 \times 10^{-5}$.

Cell segmentation

In order to find the 6-fold orientation of each cell, we proceeded to the cell segmentation of the simulated configuration. This procedure consists of the following steps. First, we define the thresholded density of the whole cell layer as

$$\Phi = \sum_{c=1}^{N_{\text{cell}}} \vartheta_H(\phi_c - \phi_{\text{th}}), \quad (10)$$

where ϑ_H is the Heaviside theta function such that $\vartheta_H(x) = 1$ if $x \geq 0$ and $\vartheta_H(x) = 0$ otherwise. Here ϕ_{th} is the threshold marking the boundary of each cell. In particular we choose $\phi_{\text{th}} = \phi_0/2$. The resulting field is $\Phi=1$ inside each cell; $\Phi = 2$ at the interface of two cells; $\Phi = 3$ or $\Phi = 4$ on the vertices. As at the interface the field ϕ_c of each cell smoothly changes from ϕ_0 to 0, therefore dropping below the threshold ϕ_{th} , it is possible to find pixels where $\Phi = 0$. These spurious features are adjusted by replacing Eq. (10) with an average over the pixels neighboring that where Φ vanishes. That is

$$\Phi = \sum_{\langle x,y \rangle} \sum_{c=1}^{N_{\text{cell}}} \frac{\vartheta_H(\phi_c)}{\max \left\{ \sum_{\langle x,y \rangle} \vartheta_H(\phi_c), 1 \right\}}, \quad (11)$$

where $\sum_{\langle x,y \rangle}$ stands for a sum over the nearest neighbors of the pixel where $\Phi = 0$. The procedure is reiterated until no change occurs in two consecutive iterations. Finally, upon segmenting the thresholded density, we identify the cell's vertices $\{\mathbf{r}_v\}_c$, as those points where $\phi_{\text{tot}} > 2$. Tissue rearrangement events are identified by tracking changes in the list of neighbors of each cell.

Cell orientation, coarse-graining and topological defects

The 6-fold orientation of a cell is computed via the shape function γ_6 defined in Eq. (2), introduced in Ref. (Armengol-Collado et al., 2023a). This construction is schematically illustrated in Fig. 4. The single-cell orientation can then be coarse-grained to construct a continuous

description of the cellular tissue (Armengol-Collado et al., 2023a). To do so, we use the shape order parameter $\Gamma_6 = \Gamma_6(\mathbf{r})$, constructed upon averaging the shape function γ_6 of the segmented cells whose center of mass, $\mathbf{r}_c$, lies within a disk of radius R centered in $\mathbf{r}$. That is

$$\Gamma_6(\mathbf{r}) = \frac{1}{N_{\text{disk}}} \sum_{c=1}^{N_{\text{cell}}} \gamma_6(\mathbf{r}_c) \mathcal{D}_H(R - |\mathbf{r} - \mathbf{r}_c|), \quad (12)$$

where $N_{\text{disk}} = \sum_c \mathcal{D}_H(R - |\mathbf{r} - \mathbf{r}_c|)$ is the number of cells whose centers lie within the disk, and the coarse-graining radius is fixed to be $R = 1.5R_{\text{cell}}$. We choose to sample the shape function on a square grid with grid-spacing equal to the nominal cell radius R_{cell} . Topological defects are then identified computing the winding number along each unit cell:

$$s = \frac{1}{2\pi} \oint_{\square} d\theta = \frac{1}{2\pi} \sum_{n=1}^4 [\theta(\mathbf{r}_{n+1}) - \theta(\mathbf{r}_n)] \bmod \frac{2\pi}{6}, \quad (13)$$

where the symbol $\square$ denotes a square unit cell in the interpolation grid and $\theta = \text{Arg}(\Gamma_6)/6$ the phase of the shape order parameter.

Nematic and Hexatic stress in MPF simulations

The combined effect of confluency and non-equilibrium cellular migration leads to the onset of internal stresses in the tissue. These can be in turn classified based on their symmetry properties with respect to the cellular shape parameter. To characterize the properties of intercellular stresses in MPF simulations, we show in **Fig. 3a,b** the longitudinal hexatic, nematic stress in the tissue. This is computed by contracting the nematic and hexatic concentration gradient tensor for each cell c ($\sigma_c^{(\text{nem})} = -k_\varphi (\nabla \phi_c)^{\otimes 2}$, $\sigma_c^{(\text{hex})} = -k_\varphi (\nabla \varphi_c)^{\otimes 6}$) with the cellular shape tensor of the corresponding order ($\mathcal{Q}_{c,p} = \mathbf{n}_{c,p}^{\otimes p}$, with $p = 2, 6$). Here, $\mathbf{n}_{c,p} = (\cos \theta_{c,p}, \sin \theta_{c,p})$ with $\theta_{c,p} = \text{Arg}(\gamma_{c,p})/p$ the angular orientation of the complex shape function γ_p relative to the c -th cell (see **Fig. 4**). Therefore, the explicit expression of the longitudinal nematic and hexatic stress is given by

$$\sigma_{\parallel}^{(\text{nem})} = \sum_{c=1}^{N_{\text{cell}}} \sigma_c^{(\text{nem})} : \mathcal{Q}_{c,2} = -k_\varphi \sum_{c=1}^{N_{\text{cell}}} \sum_{i,j} (\partial_i \varphi_c) (\partial_j \varphi_c) (n_{2,c})_i (n_{2,c})_j, \quad (14)$$

$$\sigma_{\parallel}^{(\text{hex})} = \sum_{c=1}^{N_{\text{cell}}} \sigma_c^{(\text{hex})} : \mathcal{Q}_{c,6} = -k_\varphi \sum_{c=1}^{N_{\text{cell}}} \sum_{i_1, \dots, i_6} (\partial_{i_1} \varphi_c) \dots (\partial_{i_6} \varphi_c) (n_{6,c})_{i_1} \dots (n_{6,c})_{i_6}. \quad (15)$$

Notice that positive (negative) values of the longitudinal stress correspond to contractile (extensile) stresses for both nematic and hexatic contributions.

Correlating cellular mean square displacement and defect density

The scatter plot in **Fig. 3f**, which correlates cellular mean square displacement with defect density, was constructed as follows. The system was divided into square subregions of size $\Delta \ell = 35$, each containing approximately 4 cells. For each subregion, we analyzed a time window of $\Delta t = 25 \times 10^3$ iterations, measuring both the normalized mean square displacement of cells (relative to the subregion area $\Delta \ell^2$) and the average defect density. The normalized displacement is calculated as $\text{m.s.d.} = \sum_{t=t^*}^{t^*+\Delta t} |\mathbf{r}_c(t) - \mathbf{r}_c(t-1)|^2 / \Delta \ell^2$, where t^* denotes the start time of the observation window. These measurements were collected for all subregions and time windows to generate the scatter plot.

The subregion size $\Delta \ell$ was selected to match the hexanematic crossover length-scale, ensuring each area contains about 4 cells -the fundamental unit of tissue remodeling. This size balances the need to resolve defect-rich and defect-poor regions while avoiding excessive enlargement that

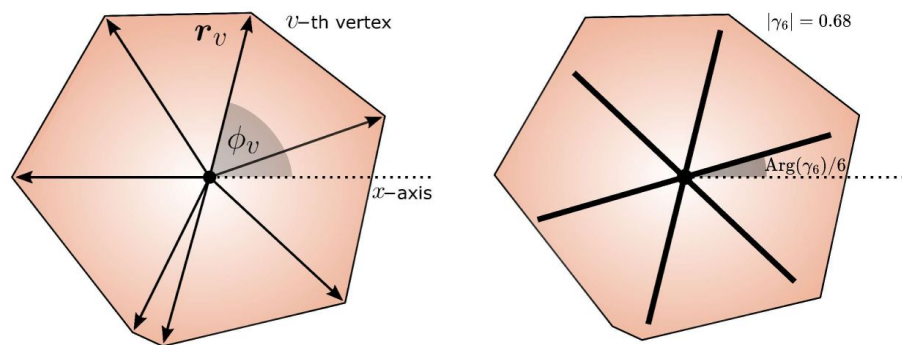


Figure 4.

Shape function.

On the left, we see a graphical representation of the 6-fold shape function γ_6 (see eq. (2) [for more details](#)) for a generic irregular polygon. On the right (black 6-legged star) the phase and magnitude of γ_6 for the same cell.

would disguise local variations. Similarly, the observation window Δt was chosen to correspond to the typical time required for a cell to traverse a subregion of size $\Delta \ell$. Shorter windows might miss remodeling events, while longer windows would average over different dynamical states.

Analysis of characteristic time of cell intercalation and T1 cycles

To analyze the temporal statistics of remodeling events we start differentiating these between T1 cycles and cell intercalation. T1 cycles are identified as events where cells first change their neighbors at time t and then they return to their initial configuration at a later time $t + T$. Analogously, for cell intercalation events-leading to a permanent tissue remodeling-we measure the time interval T between two consecutive intercalation events. The results of the analysis at varying the rotational diffusion D_r are shown in [Fig. 3g](#). For each value of D_r considered, we first identify cell intercalations and T1 cycles for each cell in the system, then we average the measured time intervals, obtaining the distribution of mean values, explicitly shown in the box plot in the inset of [Fig. 3g](#). The statistical analysis shows that the two populations (T1 cycles and intercalation events) are significantly different, with T1 cycles occurring faster than cell intercalation. We repeat this analysis at varying D_r , and we plot in [Fig. 3g](#) the characteristic time obtained as the mean value of the cell intercalations and T1 cycles distributions. Importantly we find that, while the time interval between cell intercalation events sensibly depends on the rotational diffusion, the typical timescale of T1 cycles does not.

Active flow of a hexatic defect quadrupole

Scalar order parameter

In this section we provide a derivation of [Eq. \(1\)](#). To this end, let $\Psi_6 = \Psi_6 |e^{6i\theta}$ be the hexatic complex order parameter and consider a quadrupole of $\pm 1/6$ disclinations equidistantly placed from the center of the primary cluster. The phase $\theta = \theta(\mathbf{r})$ is then given by the convolution of the average orientation in the surrounding of each defect, that is

$$\theta = -\frac{1}{6} \arctan\left(\frac{y}{x-\ell}\right) - \frac{1}{6} \arctan\left(\frac{y}{x+\ell}\right) + \frac{1}{6} \arctan\left(\frac{y+\ell}{x}\right) + \frac{1}{6} \arctan\left(\frac{y-\ell}{x}\right), \quad (16)$$

where ℓ is distance from the center. The quadrupolar distance ℓ is by definition taken to be small compared to the size of the system. Thus, expanding [Eq. \(16\)](#) for $|\mathbf{r}|/\ell \gg 1$, we obtain the simpler expression

$$\theta = -\frac{2\ell^2 \sin 2\phi}{3|\mathbf{r}|^2} + \mathcal{O}(|\ell/\mathbf{r}|^6). \quad (17)$$

Notice that the Taylor expansion features *only* the quadrupolar term of order ℓ^2 and is exact up to 6-th order in $\ell/|\mathbf{r}|$; i.e. the dipolar term, of order $\ell/|\mathbf{r}|$, and all other terms up to the 6-th order vanish identically.

This result is extremely robust, and can be derived in a number of ways. For instance, [Eq. \(16\)](#) can be obtained from the solution of the Poisson equation

$$\nabla^2 \varphi = \rho_d \quad (18)$$

where ϕ is a dual field such that $\partial_i \theta = -\epsilon_{ij} \partial_j \phi$ and the right-hand side of the **Eq. (18)** is analogous to the electrostatic charge density (Chaikin et al., 1995). At large distance from the defects, **Eq. (18)** can be solved by multipole expansion (Jackson, 1999), that is:

$$\varphi = a_0 \log \frac{r_0}{|r|} + \sum_{n=1}^{\infty} \frac{a_n \cos n\theta + b_n \sin n\theta}{|r|^n}, \quad (19)$$

where r_0 is an irrelevant length scale and a_n and b_n are coefficients given by

$$a_n = \frac{1}{n} \int dA |r|^n \cos(n\phi) \rho_d, \quad (20a)$$

$$b_n = \frac{1}{n} \int dA |r|^n \sin(n\phi) \rho_d. \quad (20b)$$

Thus, up to the quadrupole term, the expansion of ϕ is given by

$$\varphi = a_0 \log \frac{r_0}{|r|} + \frac{a_1 \cos \phi + b_1 \sin \phi}{|r|} + \frac{a_2 \cos 2\phi + b_2 \sin 2\phi}{|r|^2} + \dots \quad (21)$$

As in electrostatics, the density ρ_d is given by

$$\rho_d = \frac{1}{6} \left[-\delta(\mathbf{r} - \ell \mathbf{e}_x) - \delta(\mathbf{r} + \ell \mathbf{e}_x) + \delta(\mathbf{r} - \ell \mathbf{e}_y) + \delta(\mathbf{r} + \ell \mathbf{e}_y) \right], \quad (22)$$

where ℓ is again the distance from the center of the primary cluster. Now, because the defect quadrupole has, by construction, vanishing total strength and dipole moment, $a_0 = 0$ and $a_1 = b_1 = 0$. Of the quadrupolar terms, on the other hand, $a_2 = -\ell^2/3$ and $b_2 = 0$, thus

$$\varphi = -\frac{\ell^2 \cos 2\phi}{3|r|^2}. \quad (23)$$

Finally, going from ϕ to the original field θ one finds

$$\theta = -\frac{2\ell^2 \sin 2\phi}{3|r|^2}, \quad (24)$$

thus confirming the expression given in **Eq. (17)**.

Active Force

To shed light on the structure of the cellular flow triggered by a T1 process, we solve the Stokes equation in the presence of an active force of the form $\mathbf{f}^{(a)} = \nabla \cdot \boldsymbol{\sigma}^{(a)}$, where

$$\boldsymbol{\sigma}^{(a)} = \alpha_6 \nabla^{\otimes 4} \odot \mathbf{Q}_6, \quad (25)$$

is the active hexatic stress tensor introduced in Ref. (Armengol-Collado et al., 2023b). Calculating the divergence gives

$$\mathbf{f}^{(a)} = 960 \frac{\alpha_6 \ell^2}{|r|^7} \left\{ \left[-3 \cos 7\phi + \frac{\ell^2}{|r|^2} (3 \cos 5\phi - 14 \cos 9\phi) \right] \mathbf{e}_x + \left[3 \sin 7\phi - \frac{\ell^2}{|r|^2} (3 \sin 5\phi - 14 \sin 9\phi) \right] \mathbf{e}_y \right\}, \quad (26)$$

up to correction of order $\mathcal{O}(\ell/r)^6$. A plot of the force field is shown **Fig. 5a**.

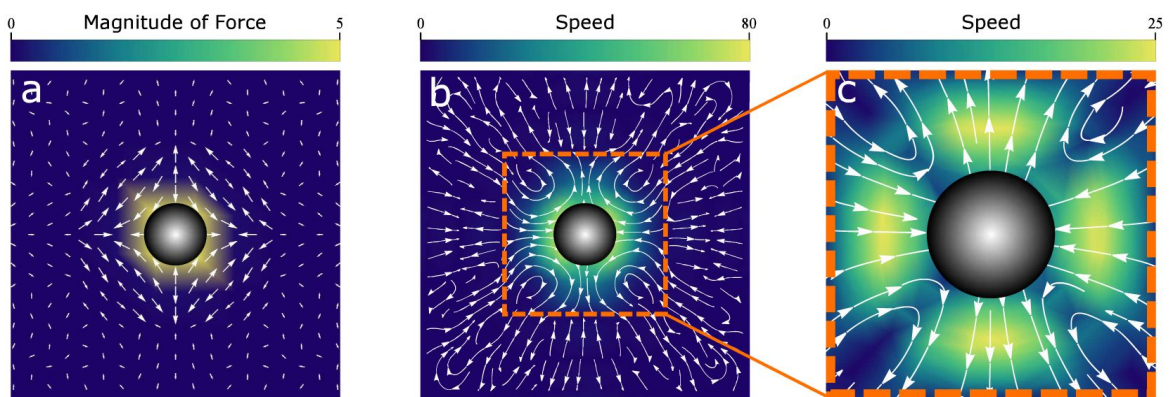


Figure 5.

Active hexatic defect quadrupole: convergent extension analytics

(a) Force field: Stream-density plot of the force field Eq. (C11). It exhibits a clear, local, convergent-extension pattern in the vicinity of the quadrupolar radius ℓ . **(b)** Velocity field: Stream density plot of the velocity field Eq. (C17). It exhibits a clear, local, convergent-extension flow pattern in the vicinity of the quadrupolar radius ℓ . **(c)** Velocity field approximated close to defect core: Stream density plot of the velocity field Eqs. (1) [\[1\]](#). It exhibits a clear, local, convergent-extension flow pattern in the vicinity of the quadrupolar radius ℓ . In all plots, the black disk corresponds to the radius of the quadrupole. Our analytical solution is valid outside the disk.

Flow field

To reconstruct the cellular motion generated by a T1 process, we solve the incompressible Stokes equation for the flow sourced the active force $\mathbf{f}^{(a)}$: i.e.

$$\eta \nabla^2 \mathbf{v} - \nabla P + \mathbf{f}^{(a)} = \mathbf{0}, \quad (27a)$$

$$\nabla \cdot \mathbf{v} = 0, \quad (27b)$$

where η is the shear viscosity and P the pressure. To this end, we return to the Oseen formal solution

$$\mathbf{v}(\mathbf{r}) = \int_0^{2\pi} d\phi' \int_{\ell}^R dr' r' \mathbf{G}(\mathbf{r} - \mathbf{r}') \cdot \mathbf{f}^{(a)}(r'), \quad (28)$$

where

$$\mathbf{G}(\mathbf{r}) = \frac{1}{4\pi\eta} \left[\left(\log \frac{\mathcal{L}}{|\mathbf{r}|} - 1 \right) \mathbb{1} + \frac{\mathbf{r} \otimes \mathbf{r}}{|\mathbf{r}|^2} \right], \quad (29)$$

is the two-dimensional Oseen tensor (see e.g. Ref. (Giomi, 2015)), with $\mathcal{L}$ a constant, and R is a large distance cut-off. Without loss of generality, one can set $\mathcal{L} = R\sqrt{e}$ in Eq. (29). To calculate the integrals in Eq. (28), we make use of the logarithmic expansion

$$\log \frac{|\mathbf{r} - \mathbf{r}'|}{\mathcal{L}} = \log \frac{r_{>}}{\mathcal{L}} - \sum_1^{\infty} \frac{1}{m} \left(\frac{r_{>}}{r_{<}} \right)^m \cos [m(\phi - \phi')], \quad (30)$$

with $r_{\leq}$ the maximum (minimum) between $|\mathbf{r}|$ and $|\mathbf{r}'|$, and of the orthogonality of trigonometric functions

$$\int_0^{2\pi} d\phi' \cos [m(\phi - \phi')] \cos n\phi' = \pi \cos n\phi \delta_{mn}. \quad (31)$$

The resulting flow field surrounding the defect quadrupole is then given by

$$\begin{aligned} \frac{\mathbf{v}}{2\alpha_6 \ell^2 / \eta} = & - \left[\frac{30 (\ell^2 - |\mathbf{r}|^2)^2}{|\mathbf{r}|^{12}} (3|\mathbf{r}|^2 \cos 8\phi + 14\ell^2 \cos 10\phi) + \frac{60}{|\mathbf{r}|^8} \cos 6\phi \left(3|\mathbf{r}|^2 + 6\ell^2 \log \frac{|\mathbf{r}|}{\ell} - 4\ell^2 \right) \right] \mathbf{r} \\ & + 6 \left(\frac{6}{|\mathbf{r}|^5} - \frac{5\ell^2}{|\mathbf{r}|^7} \right) (\cos 5\phi \mathbf{e}_x - \sin 5\phi \mathbf{e}_y) + \frac{30}{7} \left(\frac{6\ell^2}{|\mathbf{r}|^7} - \frac{7}{|\mathbf{r}|^5} \right) (\cos 7\phi \mathbf{e}_x - \sin 7\phi \mathbf{e}_y) \\ & + \frac{35}{3} \ell^2 \left(\frac{8\ell^2}{|\mathbf{r}|^9} - \frac{9}{|\mathbf{r}|^7} \right) (\cos 9\phi \mathbf{e}_x - \sin 9\phi \mathbf{e}_y). \end{aligned} \quad (32)$$

Fig. 5b shows a plot of this flow, while Fig. 5c shows a plot of the short distance approximation given in Eq. (1) of the main text.

Numerical simulations of defect annihilation and unbinding

Numerical model and validation

The time-dependent flows in **Fig. 2a(i)** and **Fig. 2b(i)** as well as the orientational field color-maps in **Fig. 2a(ii-iv)** and **Fig. 2b(ii-iv)** are obtained by numerically integrating Eqs. (3) using a vorticity-stream function finite difference scheme. All equations are discretized on a two-dimensional square grid of sizes 256×256 and 1024×1024 with periodic boundary conditions. For both grid sizes, the grid spacing is $\Delta x = \Delta y = 1$ and the time stepping $\Delta t = 0.1$. The validity of this numerical approach is benchmarked by many numerical studies on liquid crystals and active matter (see for example Refs. (Krommydas et al., 2023; Giomi et al., 2022b, 2014)). In all simulations, we set: $\rho = 1$, $\eta = 1$, $L_6 = 0.5$, $A_6 = -0.2$, $B_6 = 0.4$, $\Gamma_6 = 1$ and $\lambda_6 = 1.11$. All parameters are expressed in the arbitrary units used in the numerical simulations.

Defect annihilation and unbinding

To construct the initial configuration of Ψ_6 , we set $\ell = 7$, $|\Psi_6| = 1$ and take θ as given in Eq. (16) inside a disk of radius $R_D = 28$ and random outside. We then thermalize this configuration by keeping the orientation of the order parameter in the disk fixed, and relaxing Ψ_6 everywhere else. This allows us to obtain a defect-free configuration where $|\Psi_6| \approx 1$ everywhere, except that close to the defect cores where $|\Psi_6| \approx 0$. Notice that, on a doubly periodical domain, $\sum_i s_i = 0$. Therefore, no other topological defect is found at the end of such relaxation procedure. Simulations are carried out until the total free energy relative variation drops under 0.1% with respect to two consecutive iterations. This corresponds to a state where defects have annihilated, and the hexatic liquid crystal has achieved a smooth configuration everywhere in the simulation box. For both annihilation and unbinding numerical experiments, we scan α_6 for a wide range of positive and negative values. For any negative values of activity, we obtain increasingly sheared versions of the flow pattern in **Fig. 2a(i)**. Similarly, for positive values of α_6 we obtain increasingly sheared versions of the same flow pattern, but with the direction of the flow inverted.

Active defect dipole annihilation: the origin of the unbinding

In this section we provide a brief account of the annihilation dynamics of a pair of $\pm 1/6$ active hexatic defects, in which it is possible to recognize the fundamental mechanism driving defect unbinding. To this end, we place the defects on the x -axis at a distance of $\Delta x = 64$ and construct the initial configuration of the hexatic order parameter $\Psi_6 = e^{6i\theta}$ by setting $\theta = \pm \arctan[y/(x \pm \Delta x/2)]$ inside a disk of radius $R_D = 5$ centred at the defect cores, and random outside. We thermalize this configuration by keeping the phase of the order parameter in the two disks fixed, while relaxing Ψ_6 everywhere else. As before, this procedure allows us to obtain a state where $|\Psi_6| \approx 1$ everywhere, except that close to the two defect cores where $|\Psi_6| \approx 0$. We use this as the initial state for our annihilation experiment. Simulations are carried out until defects have annihilated and the total free energy relative variation drops under 0.1% with respect to two consecutive iterations. The model parameters, expressed in lattice units, are again: $\Delta t = 1$, $\rho = 1$, $\eta = 1$, $L_6 = 0.5$, $A_6 = -0.2$, $B_6 = 0.4$, $\Gamma_6 = 1$ and $\lambda_6 = 1.11$.

Fig. 6 shows the trajectories of the positive (red) and negative (blue) defects during annihilation, for three realizations of the activity parameter α_6 , that is $\alpha_6 = 0.1$ (contractile), $\alpha_6 = 0$ (passive) and $\alpha_6 = -0.1$ (extensile). For contractile activity, the backflow sourced by the active stress, Eq. (25), annihilation is sped up with respect to the passive case. By contrast, for extensile activity, annihilation is delayed. The same effects leads to the break up of the quadrupole into two defect pairs, provided the repulsive forces introduced by the active flow overcome the attractive Coulomb-like forces between defects.

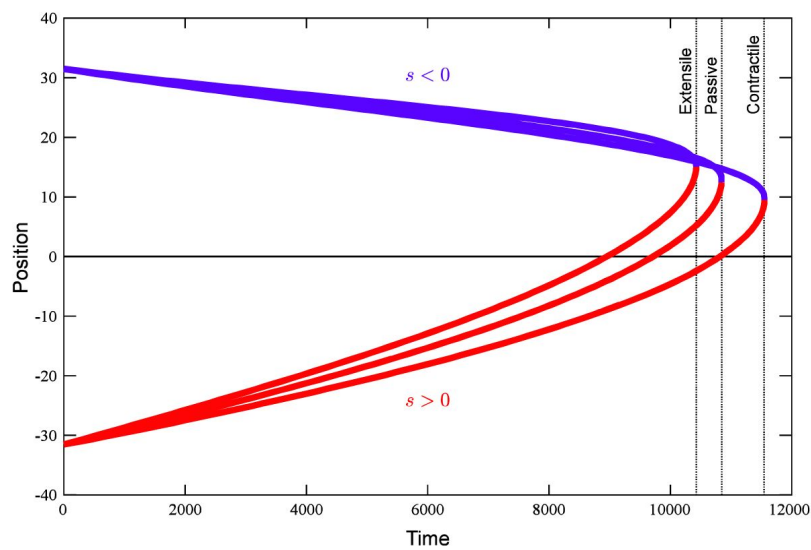


Figure 6.

Trajectories of annihilating defects in time

The red lines are the trajectories of positive and blue negative defects respectively. Defects are sped up by positive activity ($a_6 = 0.1$), but they are slowed down instead by negative activity ($a_6 = -0.1$).

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Reviewer #1 (Public review):

Summary:

This paper investigates the physical mechanisms underlying cell intercalation, which then enables collective cell flows in confluent epithelia. The authors show that T1 transitions (the topological transitions responsible for cell intercalation) correspond to the unbinding of groups of hexatic topological defects. Defect unbinding, and hence cell intercalation and collective cell flows, are possible when active stresses in the tissue are extensile. This result helps to rationalize the observation that many epithelial cell layers have been found to exhibit extensile active nematic behavior.

Strengths:

The authors obtain their results based on a combination of active hexanematic hydrodynamics and a multiphase field (MPF) model for epithelial layers, whose connection is a strength of the paper. With the hydrodynamic approach, the authors find the active flow fields produced around hexatic topological defects, which can drive defect unbinding. Using the MPF simulations, the authors show that T1 transitions tend to localize close to hexatic topological defects.

<https://doi.org/10.7554/eLife.105397.2.sa2>

Reviewer #2 (Public review):

Summary:

This paper studies the role of hexatic defects in the collective migration of epithelia. The authors emphasize that epithelial migration is driven by cell intercalation events and not just isolated T1 events, and analyze this through the lens of hexatic topological defects. Finally, the authors study the effect of active and passive forces on the dynamics of hexatic defects using analytical results, and numerical results in both continuum and phase-field models. The results are very interesting, and highlight new ways of studying epithelial cell migration through the analysis of the binding and unbinding of hexatic defects.

Strengths:

(1) The authors convincingly argue that intercalation events are responsible for collective cell migration, and that these events are accompanied by the formation and unbinding of hexatic topological defects. (2) The authors clearly explain the dynamics of hexatic defects during T1 transitions, and demonstrate the importance of active and passive forces during cell migration. (3) The paper thoroughly studies the T1 transition through the viewpoint of hexatic defects. A continuum model approach to study T1 transitions in cell layers is novel and can lead to valuable new insights.

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Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

This paper investigates the physical mechanisms underlying cell intercalation, which then enables collective cell flows in confluent epithelia. The authors show that T1 transitions (the topological transitions responsible for cell intercalation) correspond to the unbinding of groups of hexatic topological defects. Defect unbinding, and hence cell intercalation and collective cell flows, are possible when active stresses in the tissue are extensile. This result helps to rationalize the observation that many epithelial cell layers have been found to exhibit extensile active nematic behavior.

Strengths

The authors obtain their results based on a combination of active hexanematic hydrodynamics and a multiphase field (MPF) model for epithelial layers, whose connection is a strength of the paper. With the hydrodynamic approach, the authors find the active flow fields produced around hexatic topological defects, which can drive defect unbinding. Using the MPF simulations, the authors show that T1 transitions tend to localize close to hexatic topological defects.

We are grateful to Reviewer #1, for appreciating and highlighting the strengths of work.

Weaknesses

Citations are sometimes not comprehensive. Cases of contractile behavior found in collective cell flows, which would seemingly contradict some of the authors' conclusions, are not discussed.

I encourage the authors to address the comments and questions below.

We are thankful to Reviewer #1, for their questions and comments. We have addressed them point by point below, and have amended the manuscript accordingly.

(1) In Equation 1, what do the authors mean by the cluster's size ℓ ? How is this quantity defined? The calculations in the Methods suggest that ℓ indicates the distance between the p-atic defects and the center of the T1 cell cluster, but this is not clearly defined.

We are thank Reviewer #1 for their question. We define the cluster size as the initial distance between the center of the quadrupole and any defect (see Methods). In a primary cell cluster, where cells themselves are the defects, the cluster's size is the distance between the center of the central junction and the center of any cell in the cluster. Hence, this is half the diameter of an cell which, for example in a typical, confluent MDCK epithelial monolayer, would be about 10 μ m. We have added this clarification in the definition of the cluster size, above Eq. (1).

(2) The multiphase field model was developed and reviewed already, before the Loewe et al. 2020 paper that the authors cite. Earlier papers include Camley et al. PNAS 2014, Palmieri et al. Sci. Rep. 2015, Mueller et al. PRL 2019, and Peyret et al. Biophys. J. 2019, as reviewed in Alert and Trepot. Annu. Rev. Condens. Matter Phys. 2020.

We thank the referee for their suggestion to incorporate further MPF literature. We have done so in the amended manuscript.

(3) At what time lag is the mean-squared displacement in Figure 3f calculated? How does the choice of a lag time affect these data and the resulting conclusions?

The scatter plot in Fig. 3f was constructed by dividing the system into square subregions of size $\Delta\ell = 35$ l.u., each containing approximately 4 cells. For each subregion, we analyzed a time window of $\Delta t = 25 \times 10^3$ iterations, measuring both the normalized mean square displacement of cells (relative to the subregion area $\Delta\ell^2$) and the average defect density. The

normalized displacement is calculated as m.s.d. $= \sum_{t=t^*}^{t^*+\Delta t} |r_c(t) - r_c(t-1)|^2 / \Delta\ell^2$, where t^* denotes the

start time of the observation window. We chose the time window Δt used to compute the mean square displacement to match the characteristic duration of T1 events and defect lifetimes in our simulations. Observation times much longer ($\Delta t > 35 \times 10^3$) than the typical T1 event duration would cause the two sets of data points to merge into a single group, suggesting no correlation between cell motility and defect density beyond defect life-time.

(4) The authors argue that their results provide an explanation for the extensile behavior of cell layers. However, there are also examples of contractile behavior, such as in Duclos et al., Nat. Phys., 2017 and in Pérez-González et al., Nat. Phys., 2019. In both cases, collective cell flows were observed, which in principle require cell intercalations. How would these observations be rationalized with the theory proposed in this paper? Can these experiments and the theory be reconciled?

The contractile or extensile nature of stress in epithelia depends crucially on the specific tissue type and its biological context. Different cell populations, depending on their position along the epithelial/mesenchymal spectrum, can exhibit either contractile or extensile behaviors. Our theory applies to tissues where hexatic order dominates at the cellular scale, particularly in confluent systems where neighbor exchanges occur primarily through T1 transitions. In contrast, the systems studied by Duclos et al., Nat. Phys. (2018) and Perez-Gonzalez et al. (Nat. Phys., 2019) exhibit nematic order at the cellular level, meaning their dynamics are governed by fundamentally different mechanisms. Since our framework is derived for hexatic-dominated tissues, it does not directly apply to those cases, though a hybrid hexanematic descriptions previously developed by some of the authors in Armengol-Collado et al. eLife 13:e86400 (2024) could help reconcile these observations. In general, a key distinction must be made between the contractility of individual cells and the extensile/contractile nature of the collective force network. To illustrate this, consider a cell exerting a 6-fold symmetric force distribution: each vertex force arises from an imbalance in junctional tensions with neighboring cells, which are themselves contractile due to actomyosin activity. However, the resulting vertex forces can be either contractile or extensile depending on network geometry and tension distribution. This is captured in our coarse-grained description [see Armengol-Collado et al. eLife 13:e86400 (2024)], where the active stress emerges from higher-order moments of cellular forces. Specifically, the deviatoric part of the hexatic active stress tensor, where r is the cell radius, the number cell density and the intensity of cellular tension. The negative sign of the coefficient of the active stress shows that the active stress is extensile—consistently with observations in various epithelial systems (e.g., Saw et al., Nature 2017; Blanch-Mercader et al., Phys. Rev. Lett. 2018). Finally, we note that the connection between cellular-scale forces and large-scale extensility has been rationalized in other contexts, such as active nematics (Balasubramaniam et al., Nat. Mater. 2021).

Reviewer #2 (Public Review):

This paper studies the role of hexatic defects in the collective migration of epithelia. The authors emphasize that epithelial migration is driven by cell intercalation events and not just isolated T1 events, and analyze this through the lens of hexatic topological defects. Finally, the authors study the effect of active and passive forces on the dynamics of

hexatic defects using analytical results, and numerical results in both continuum and phase-field models.

The results are very interesting and highlight new ways of studying epithelial cell migration through the analysis of the binding and unbinding of hexatic defects.

We are grateful to Reviewer #2, for their interest and for emphasizing the novelty of our work.

Strengths

(1) The authors convincingly argue that intercalation events are responsible for collective cell migration, and that these events are accompanied by the formation and unbinding of hexatic topological defects.

(2) The authors clearly explain the dynamics of hexatic defects during T1 transitions, and demonstrate the importance of active and passive forces during cell migration.

(3) The paper thoroughly studies the T1 transition through the viewpoint of hexatic defects. A continuum model approach to study T1 transitions in cell layers is novel and can lead to valuable new insights.

We thank the Reviewer for their kind and supporting words, and for highlighting the clarity, persuasiveness, and thoroughness.

Weaknesses

(1) The authors could expand on the dynamics of existing hexatic defects during epithelial cell migration, in addition to how they are created during T1 transitions.

We thank the referee for their comment. The detailed analysis of dislocation-pair unbinding modes and their statistical impact on the transition to collective migration is comprehensively addressed in our subsequent work Puggioni et al., arXiv:2502.09554. In the present study, we focus specifically on the fundamental mechanism enabling dislocation unbinding: active extensile stresses generate flows that drive dislocation pairs apart, while passive elastic stresses tend to pull them together (Krommydas et al., Phys. Rev. Lett. 2023; Armengol- Collado et al., arXiv:2502.13104). When active forces dominate over passive restoring forces, the dislocations unbind. This represents a crucial distinction from classical Berezinskii–Kosterlitz–Thouless or Kosterlitz–Thouless–Halperin–Nelson–Youn transitions, where thermal fluctuations drive defect unbinding. In our system, the process is fundamentally activity-driven. Nevertheless, the resulting state - characterized by unbound defects and collective migration - bears strong analogy to the melting transition in equilibrium systems. We emphasize that the dynamics of passive defects has been previously examined in Krommydas et al., Phys. Rev. Lett. 2023. A discussion of these aspects can be found in the Appendix “Numerical simulations of defect annihilation and unbinding”.

(2) The different terms in the MPF model used to study cell layer dynamics are not fully justified. In particular, it is not clear why the model includes self-propulsion and rotational diffusion in addition to nematic and hexatic stresses, and how these quantities are related to each other.

We thank the referee for their comment. The MPF model’s terms (e.g., self-propulsion, rotational diffusion), reflect the stochastic, deformable nature of cells as active droplets migrating with near-constant speed. We emphasize that self-propulsion is the only non-equilibrium mechanism in our model — no additional active stresses (nematic or hexatic) are

imposed. We have clarified this point in the revised manuscript and expanded our discussion of the MPF model.

(3) The authors could provide some physical intuition on what an active extensile or contractile term in the hexatic order parameter means, and how this is related to extensility and contractility in active nematics and/or for cell layers.

We thank the referee for their comment. As we explain in the reply to comment [4] of Reviewer #1, the contractile or extensile nature of stress in epithelia depends crucially on the specific tissue type and its biological context. Different cell populations, depending on their position along the epithelial/mesenchymal spectrum, can exhibit either contractile or extensile behaviors. Our theory applies to tissues where hexatic order dominates at the cellular scale, particularly in confluent systems where neighbor exchanges occur primarily through T1 transitions. In contrast, the systems studied by Duclos et al., Nat. Phys. (2018) and Perez-Gonzalez et al. (Nat. Phys., 2019) exhibit nematic order at the cellular level, meaning their dynamics are governed by fundamentally different mechanisms. Since our framework is derived for hexatic-dominated tissues, it does not directly apply to those cases, though a hybrid hexanematic descriptions previously developed by some of the authors in Armengol-Collado et al. eLife 13:e86400 (2024) could help reconcile these observations. In general, a key distinction must be made between the contractility of individual cells and the extensile/contractile nature of the collective force network. To illustrate this, consider a cell exerting a 6-fold symmetric force distribution: each vertex force arises from an imbalance in junctional tensions with neighboring cells, which are themselves contractile due to actomyosin activity. However, the resulting vertex forces can be either contractile or extensile depending on network geometry and tension distribution. This is captured in our coarse-grained description [see Armengol-Collado et al. eLife 13:e86400 (2024)], where the active stress emerges from higher-order moments of cellular forces. Specifically, the deviatoric part of the hexatic active stress tensor, where r is the cell radius, the number cell density and the intensity of cellular tension. The negative sign of the coefficient of the active stress shows that the active stress is extensile—consistently with observations in various epithelial systems (e.g., Saw et al., Nature 2017; Blanch-Mercader et al., Phys. Rev. Lett. 2018). Finally, we note that the connection between cellular-scale forces and large-scale extensility has been rationalized in other contexts, such as active nematics (Balasubramaniam et al., Nat. Mater. 2021).

Recommendations for the Authors: Reviewer #2 (Recommendations for the Authors):

(1) The authors point out that hexatic topological defects are produced in quadrupoles (L109). Does this also mean that these defects can be annihilated only in quadrupoles as well? In the same vein, are hexatic defects always bound in pairs, as suggested by the schematics, or is it possible to observe an isolated hexatic defect?

We thank the referee for their question. Hexatic disclinations (the defect monopoles discussed in this work), much like electrons and positrons, can annihilate in any number of neutral charge configuration (dipole, quadrupole, octupole, etc.). Unbinding a pair of hexatic disclination, however, costs much more energy than unbinding a quadrupole to dipoles. Hence isolated defects appear in abundance only in late, fully disordered phase, where the system has completely “melted”. For more details on how defect unbinding modes affect tissue dynamics, please see our subsequent work Puggioni et al., arXiv:2502.09554.

(2) Could you clarify if the flows described in Figures 2(a)-(b), panel (i) are driven by a passive backflow term without activity? Could you compare the magnitudes of these flows compared to the typical active terms?

We thank the referee for their question. In panel 2(b) there is only passive backflow. In 2(a) instead, both terms are included, and are in a regime of parameters where the active flow overcomes the passive flow (and hence the active force overcomes the passive force as delineated in the discussions section). In turn, the magnitude of the passive flows, is studied in detail in our previous work Krommydas et al., (Phys. Rev. Lett. 2023).

(3) Could you clarify how the continuum hexatic model and MPF model are related to each other? What are the similarities and differences in the dynamics of these models?

We thank the referee for this insightful question. A key point of our work is precisely that the continuum hexatic model and the MPF (Multi-Phase Field) model are distinct in nature.

The MPF model is an established agent-based framework used to simulate tissue dynamics at the cellular level. It captures individual cell behaviors and interactions through phase-field variables. In our work, we use the MPF model as a benchmark to extract statistical features of tissue dynamics, such as defect motion and orientational correlations. In contrast, our continuum hexatic model is a coarse-grained hydrodynamic theory that describes the dynamics of orientational order in active tissues. It is built on symmetry principles and conservation laws, and it does not rely on microscopic cell-level details. Instead, it captures the collective behavior of the system through a hexatic order parameter and its coupling to flow and activity.

Despite their conceptual differences, the MPF model and our hydrodynamic theory exhibit similar statistical features. This agreement—also observed in the independent study by Jain et al. (Phys. Rev. Res. 2024)—provides strong support for the validity and generality of our continuum description.

(4) When multiple references by the same author and year are cited using alphabets, the second alphabet is not in bold e.g. Giomi et al., 2022b, a in Line 75, and others.

We are grateful to the referee carefully going through the manuscript and pointing out these typos. We have corrected them in the amended manuscript.

Reviewer #3 (Public Review):

In this manuscript, the authors discuss epithelial tissue fluidity from a theoretical perspective. They focus on the description of topological transitions whereby cells change neighbors (T1 transitions). They explain how such transitions can be described by following the fate of hexatic defects. They first focus on a single T1 transition and the surrounding cells using a hydrodynamic model of active hexatics. They show that successful T1 intercalations, which promote tissue fluidity, require a sufficiently large extensile hexatic activity in the neighborhood of the cells attempting a T1 transition. If such activity is contractile or not sufficiently extensile, the T1 is reversed, hexatic defects annihilate, and the epithelial network configuration is unchanged. They then describe a large epithelium, using a phase field model to describe cells. They show a correlation between T1 events and hexatic defects unbinding, and identify two populations of T1 cells: one performing T1 cycles (failed T1), and not contributing to tissue migration, and one performing T1 intercalation (successful T1) and leading to the collective cell migration.

Strengths

The manuscript is scientifically sound, and the variety of numerical and analytical tools they use is impressive. The approach and results are very interesting and highlight the relevance of hexatic order parameters and their defects in describing tissue dynamics.

We thank the Reviewer for recognizing the scientific soundness of the manuscript, the breadth of numerical and analytical tools employed, as well as their interest in our work.

Weaknesses

(1) Goal and message of the paper. (a) In my opinion, the article is mainly theoretical and should be presented as such. For instance, their conclusions and the consequences of their analysis in terms of biology are not extremely convincing, although they would be sufficient for a theory paper oriented to physicists or biophysicists. The choice of journal and potential readership should be considered, and I am wondering whether the paper structure should be re-organized, in order to have side-by-side the methods and the results, for instance (see also below).

We thank the referee for their criticism. In response, we have made an effort to reword certain parts of the manuscript. As with any theoretical study, the biological implications of our work can only be fully assessed through experimental validation — a prospect we look forward to. Nevertheless, we have submitted our work to the subsection of Physics of Life, which we believe is perfectly suited to our content.

(b) Currently, the two main results sections are somewhat disconnected, because they use different numerical models, and because the second section only marginally uses the results from the first section to identify/distinguish T1.

We thank the referee, for their comment. In the second section we are using statistics from the MPF model, to support the analytical and numerical findings of our hydrodynamic theory of cell intercalation. In the time between our submission, further qualitative evidence have been brought to light in the work of Jain et al. (Phys. Rev. Res. 2024).

(2) Quite surprisingly, the authors use a cell-based model to describe the macroscopic tissuescale behavior, and a hydrodynamic model to describe the cell-based events. In particular, their hydrodynamic description (the active hexatic model) is supposed to be a coarse-grained description, valid to capture the mesoscopic physics, and yet, they use it to describe cellscale events (T1 transitions). For instance, what is the meaning of the velocity field they are discussing in Figure 2? This makes me question the validity of the results of their first part.

We thank the referee for their comment. There are many excellent discrete models of epithelial tissues in the literature (e.g., Bi et al., Phys. Rev. X 2016; Pasupalak et al., Soft Matter 2020; Graner et al., Phys. Rev. Lett. 1992), each capturing essential biological features such as cell division, apoptosis and sorting. While these models have provided invaluable insights, our work takes a different approach by developing a continuum theory aimed at describing epithelial dynamics at two levels: (1) mesoscopic intercalation events and (2) macroscopic collective migration. Crucially, our goal is not to replicate a specific discrete model — which would risk constructing a “model of a model” — but rather to derive a hydrodynamic description of tissue dynamics grounded in symmetry principles and conservation laws. Along this logic, the velocity field in our theory should be interpreted as an Eulerian (continuum) velocity, representing the coarse-grained flow of the tissue rather than the Lagrangian motion of individual cells. This distinction is central to our framework, which operates at scales where cellular details are averaged out, yet retains the essential physics of hexatic order and active stresses. We validate our predictions against the Multiphase Field (MPF) model. [We thank Reviewer 1 for their suggestion to incorporate further MPF literature.] Furthermore, Jain et al. (Phys. Rev. Res. 2024) have used the MPF to predict flow patterns around T1 transitions and obtained results compatible with those of our hydrodynamic theory. From this comparison we can conclude that both the MPF and our

theory are able to capture the same aspect of cell intercalation in epithelial layer. This, however, does not imply that other discrete models of epithelia can reproduce this aspect too, nor that our theory is specifically tailored to the MPF model. We have clarified these points in the revised manuscript and expanded our discussion of the MPF model.

(3) The quality of the numerical results presented in the second part (phase field model) could be improved. (a) In terms of analysis of the defects. It seems that they have all the tools to compare their cell-resolved simulations and their predictions about how a T1 event translates into defects unbinding. However, their analysis in Figure 3e is relatively minimal: it shows a correlation between T1 cells and defects. But it says nothing about the structure and evolution of the defects, which, according to their first section, should be quite precise.

We thank the referee for their comment. Further qualitative evidence have been brought to light in the work of Jain et al. (Phys. Rev. Res. 2024), where the exact flow pattern predicted by our hydrodynamic theory is obtained, in the MPF, around cells undergoing T1 rearrangements.

(b) In terms of clarity of the presentation. For instance, in Figure 3f, they plot the mean-square displacement as a function of a defect density. I thought that MSD was a time-dependent quantity: they must therefore consider MSD at a given time, or averaged over time. They should be explicit about what their definition of this quantity is.

We thank the referee for raising this point. As clarified in our response to Reviewer 1, point 3, the mean square displacement (MSD) plotted in Fig. 3f is computed over a fixed time window of $\Delta t = 25 \times 10^3$ iterations, chosen to match the typical duration of T1 events and defect lifetimes. [See also reply to Reviewer #1, point (3).] The MSD is normalized by the subregion area and averaged over time within each window. We have now made this explicit in the amended version of the manuscript.

(c) In terms of statistics. For instance, Figure 3g is used to study the role of rotational diffusion on the average time between T1s. The error bars in this figure are huge and make their claims hardly supported. Their claim of a “monotonic decay” of the average time between intercalations is also not fully supported given their statistics.

We appreciate the Reviewer’s comment regarding the statistical robustness of Fig. 3g. While we acknowledge that the error bars are substantial – reflecting the inherent variability in cell intercalation dynamics – the yellow curve does exhibit a consistent downward trend in the average time between T1 transitions as rotational diffusion increases. This monotonic decrease is visible across the entire range of variation of the rotational diffusion D_r , and is statistically supported when considering the trend over independent simulations. To address this concern, we have revised the main text to adjusted the wording: instead of stating that “the former is a monotonically decreasing function of D_r ,” we now write that “the former displays a decreasing trend with D_r ,” which better reflects the statistical variability while preserving the observed behavior.

Reviewer #3 (Recommendations for the Authors):

(1) Section 1 is difficult to follow due to multiple reasons: early but delayed definitions, unclear use of T1 intercalation vs. T1 cycles, disconnected figures and unclear simulation descriptions. We recommend including simulation setup details earlier and restructuring the flow of arguments.

We thank the referee for their comment. We have made an effort in rewording and clarifying things in our amended manuscript. We are slightly confused by what they mean by “early but delayed definitions”, if they could clarify, we would be happy to amend the position and phrasing of these definitions accordingly.

(2) It could be useful to have an additional figure early on defining schematically hexatic defects and an illustration showing an epithelium (or a simulation), similar to what the authors have produced in some of their other publications on this topic.

We thank the referee for their comment. Figures 3c and 3d show what a hexatic defect looks like in a simulation of the epithelium. Following the referee’s recommendation, we have added a note in the caption of figure 3, citing our work where we show the same defects in MDCK epithelial monolayers (Armengol et al., Nat. Phys. 2023).

(3) Minor points and typos:

Line 88: the bond between vertices shrinks, not the vertices.

Figure 1: the $1/6$ is displayed as 1 6 (fraction bar missing).

Line 232: “and order” → “one/an order”.

Line 237: Fig. 3g) → Fig. 3g

Line 298: “nu” and “v” hard to distinguish in eLife font.

Methods: define all notation clearly (e.g., tensor product exponent, D/Dt in Eq. 3c).

Methods: “cell orientation, coarse-graining and topological defects” section is difficult to follow, schematic would help.

Line 457 onward: unclear how panels (ii-iv) of Fig. 2ab are obtained.

Line 480 onward: not referenced in main text.

Figure 2: “avalancHe” typo.

Figure 2 caption: “cell intercalatION” typo.

Movies are neither referenced nor explained.

Figure 5 and 6 are not referenced in the main text.

We thank the referee for their detailed read of the paper. We have corrected all typos.

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